Semantic Movie Recommendation System using Vector Database

GenAI EX-6.ipynb

AIM:

Implement a scalable semantic search engine that utilizes high-dimensional vector embeddings and a vector database (FAISS) to retrieve contextually relevant media from a large-scale dataset.

CODE:

```
# 🔹 Install Libraries
!pip install pandas numpy faiss-cpu sentence-transformers

# 🔹 Imports
import pandas as pd
import numpy as np
import faiss
from sentence_transformers import SentenceTransformer
from google.colab import files

# 🔹 Upload Dataset
uploaded = files.upload()
movies = pd.read_csv('movies.csv')

# 🔹 Preprocessing + Feature Engineering (COMBINED)
movies = movies.dropna()
movies['genres'] = movies['genres'].str.replace('|', ' ')
movies['combined'] = movies['title'] + " " + movies['genres']

# 🔹 Load Embedding Model + Generate Embeddings
model = SentenceTransformer('all-MiniLM-L6-v2')
embeddings = model.encode(movies['combined'].tolist(),
show_progress_bar=True)

# 🔹 Create FAISS Index (Cosine Similarity)
dimension = embeddings.shape[1]
faiss.normalize_L2(embeddings)
```

```python
index = faiss.IndexFlatIP(dimension)
index.add(embeddings)

# 🔹 Store Metadata
metadata = movies[['movieId', 'title', 'genres']]

print("Total Movies Indexed:", index.ntotal)

# 🔹 User Query + Search (COMBINED)
query = input("Enter your movie search: ")

query_embedding = model.encode([query])
faiss.normalize_L2(query_embedding)

k = 5
scores, indices = index.search(query_embedding, k)

# 🔹 Display Results
print("\n🎬 Top Semantic Recommendations:\n")
for i, idx in enumerate(indices[0]):
    print(f"{i+1}. {metadata.iloc[idx]['title']}
({metadata.iloc[idx]['genres']})")
    print(f"   Similarity Score: {scores[0][i]:.4f}\n")

# 🔹 Keyword Search Comparison
print("\n🔍 Keyword Search Results:\n")
keyword_results = movies[movies['combined'].str.contains(query,
case=False, na=False)]
print(keyword_results['title'].head(5))
```

OUTPUT:

```
Total Movies Indexed: 9742
Enter your movie search: shrek

🎬 Top Semantic Recommendations:

1. Shrek (2001) (Adventure Animation Children Comedy Fantasy Romance)
   Similarity Score: 0.6625

2. Shrek the Third (2007) (Adventure Animation Children Comedy Fantasy)
   Similarity Score: 0.6552

3. Shrek the Halls (2007) (Adventure Animation Comedy Fantasy)
   Similarity Score: 0.6548

4. Shrek 2 (2004) (Adventure Animation Children Comedy Musical Romance)
   Similarity Score: 0.6505

5. Shrek Forever After (a.k.a. Shrek: The Final Chapter) (2010) (Adventure Animation Children Comedy Fantasy IMAX)
   Similarity Score: 0.6104


🔍 Keyword Search Results:

3194                                        Shrek (2001)
5160                                      Shrek 2 (2004)
6486                               Shrek the Third (2007)
6915                               Shrek the Halls (2007)
7360    Shrek Forever After (a.k.a. Shrek: The Final C...
Name: title, dtype: object
```